16 April 1954
SUPERSEDING
MIL-F-20413
3 December 1951

MILITARY SPECIFICATION

FUZE, POINT INITIATING, M90A1
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers loading, assembling and packing of the M90A1 point initiating fuze for use with artillery ammunition.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications, and all drawings, specifications and standards referenced thereon, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-G-2550 -General Specification for Ammunition Except Small Arms Ammunition

STANDARDS

MILITARY

MIL-STD-350 -Jolt Test For Use In Production of Fuzes

DRAWINGS

ORDNANCE CORPS

73-2-236 -Fuze, Point Initiating, M90A1, Assembly
81-3-35 -Machine, Jumble Testing
81-3-37 -Fixture For Jumbling Fuzes. Assembly and Details
81-3-102 -Fixture Testing Detonator Sensitivity Details

PUBLICATIONS

ORDNANCE CORPS

ORD-SIP-M17-14-Fuze, PI, M90A1, Loaded

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(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Material and parts shall be in accordance with applicable drawings and specifications.

3.2. The fuzes shall comply with all requirements specified on Drawing 72-2-236 and all drawings referenced thereon, and with all requirements specified in applicable specifications.

3.3 Jolt and jumble.-The fuze shall withstand the jolt test specified in Standard MIL-STD-350 and after completion of the jolt test, the fuze used in the jolt test shall withstand 3600 complete revolutions in the jumble machine without having any of the following malfunctions when tested as specified in 4.5.

a. explosive elements explode,
b. fuzes come apart,
c. fuzes partially or fully arm,
d. inner parts, including explosive elements broken, deformed or displaced in such a manner as to make the fuze unsafe to handle or dangerous to fire.

3.4 Static functioning.-The fuzes shall function satisfactorily, penetrating a 1/8 inch steel plate, when fired statically as specified in 4.6.

3.5 Proving ground requirements for the fuze.

3.5.1 There shall be no premature burst when tested as specified in 4.7. A premature burst shall be defined as a functioning of the fuze in the rifle or in flight before impact.

3.5.2 Not more than one fuze shall fail to detonate the projectile filler satisfactorily when tested as specified in 4.7.

3.6 Workmanship.-All parts shall be free from chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot.-A lot shall consist of all fuzes manufactured in one unchanged process in accordance with the same drawing, same drawing revision and same specification, same specification revision. Each lot shall contain:

a. parts from not more than one interfix lot number of one manufacturer,
b. detonators from not more than two lots,
c. primers from not more than two lots,
d. auxiliary boosters from not more than two lots.

4.2 Sampling.-Number of samples, acceptance and rejection criteria used for determination of lot acceptance for tests as specified in 4.4, 4.5 and 4.6 shall be in accordance with Standard Inspection Procedure ORD-SIP-M17-14.

4.3 Inspection.-Inspection shall be as specified in Specification MIL-G-2550.

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4.4 Rotor and lock assembly test.-Before assembling the primer assembly and after staking the body pin, each fuze in the lot, less ogive, auxiliary booster holder, auxiliary booster assembly and disc, shall be rotated in a fixture as shown in figure 1. The axis of the spindle of the spinning machine shall be vertical and the spindle shall be of the intermittently operating type, that is, shall be stopped for inserting and removing the parts. The spindle shall be accurately aligned and balanced, and the spinning machine shall be rigidly mounted on a solid base. The fuze shall arm (that is the detonator shall align itself along the axis of the fuze) at the arming speed specified on the drawing, and shall not arm when spun at the non-arming speed specified on the drawing. However, at no time during the arming test shall the speed of rotation of the spindle exceed by more than 2 percent the arming speed specified on the drawing and at no time during the non-arming test shall the speed of rotation of the spindle be more than 2 percent less than the non-arming speed specified on the drawing.

4.5 Jolt and jumble test.-Sample fuzes selected from each lot shall be subjected to test as specified in MIL-STD-350. After completion of the jolt test, the sample fuzes used in the jolt test shall be subjected to a jumble test. Assemble the fuzes to an adapter (figure 2) and place one at a time in Jumble Machine, Drawing 81-3-35. Give the machine 3600 complete revolutions at the speed specified on Drawing 81-3-35. Examine the sample fuzes upon completion of this test to determine compliance with 3.3.

4.6 Static functioning test.-Sample fuzes shall be selected and the ogives and lock assemblies shall be removed. The sample fuzes shall be placed on a steel plate and arranged for firing, with the rotor armed and secured in position with a modified lock as shown in Figure 3. The samples shall be fired by allowing a one pound weight to drop one foot, striking the firing pin. Each sample fuze shall be observed to determine compliance with 3.4. (Note.-The samples used in the test shall be handled carefully to prevent accidental prematures. Explosive tests shall be conducted within or behind a suitable barricade.)

4.7 Proving ground test.

4.7.1 Procedure.-Twenty sample fuzes shall be assembled in 57-mm HEAT Shell, M307, and assembled to complete rounds using a propelling charge to give standard service velocity. The complete rounds shall be fired from a 57-mm Rifle, M18, against a wood bursting screen 2 inches thick, placed not less than 300 feet from the muzzle of the weapon. Observe for functioning to determine compliance with 3.5.1 and 3.5.2.

4.7.2 Acceptance.-The lot shall be considered acceptable provided no premature bursts occurred and provided at least 19 of the 20 fuzes detonate the projectile filler satisfactorily. The lot shall be considered eligible for retest under the provisions of 4.7.3 provided no premature burst occurs and if 18 of the 20 fuzes detonate the projectile filler satisfactorily. The lot shall be rejected if a premature burst occurs or if less than 18 of the 20 fuzes detonate the projectile filler satisfactorily.

4.7.3 Retest.-A retest as specified in 4.7.2 may be made if requested by the contractor, using 40 fuzes. The lot shall be considered acceptable on retest provided that no premature burst occurred and at least 39 of the 40 fuzes detonate the projectile filler satisfactorily.

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5. PREPARATION FOR DELIVERY

5.1 Packing, labeling and marking.-Packing, labeling and marking shall be as specified on the applicable drawing.

5.2 Data cards.-Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

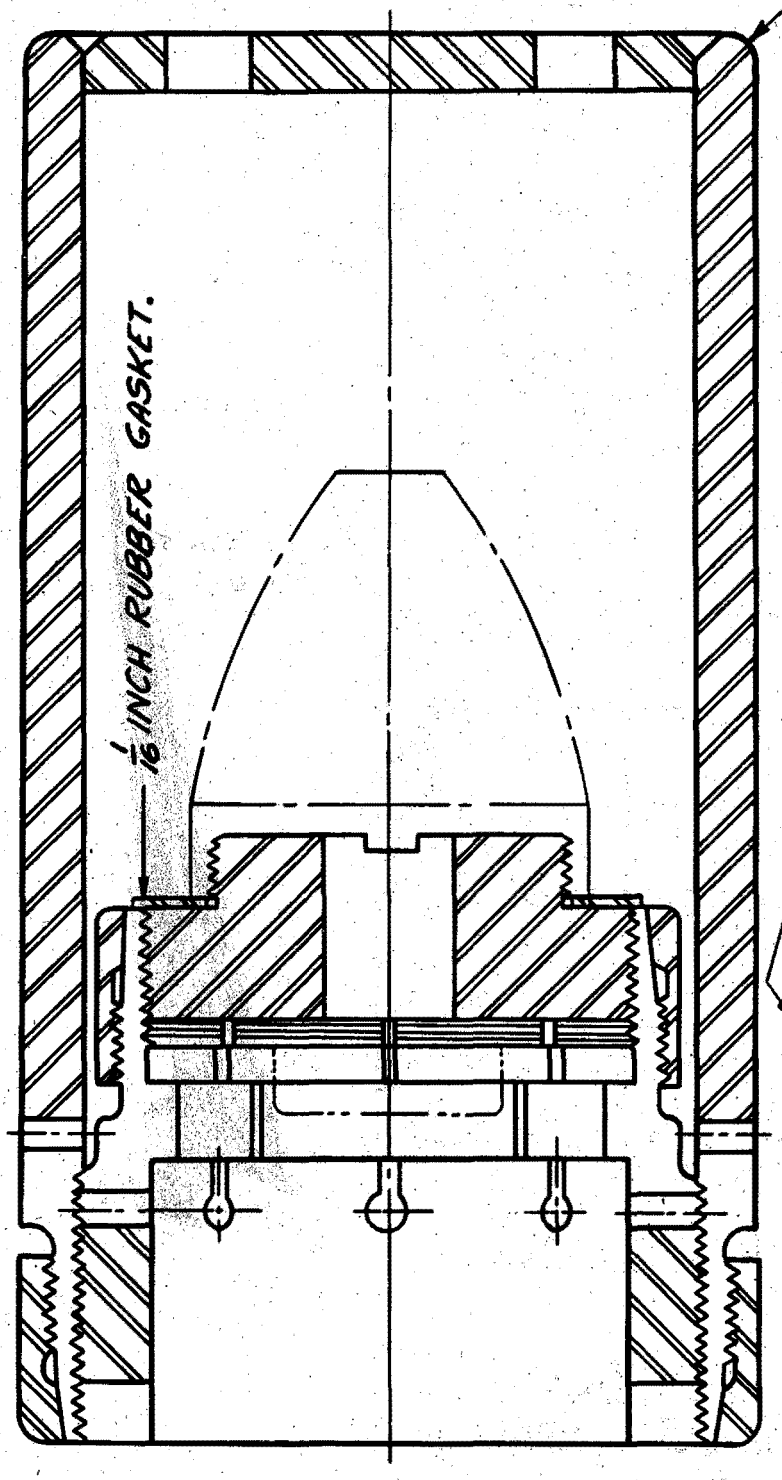
6.1 Ordering data. - Procurement documents should specify the title, number, and date of this specification.

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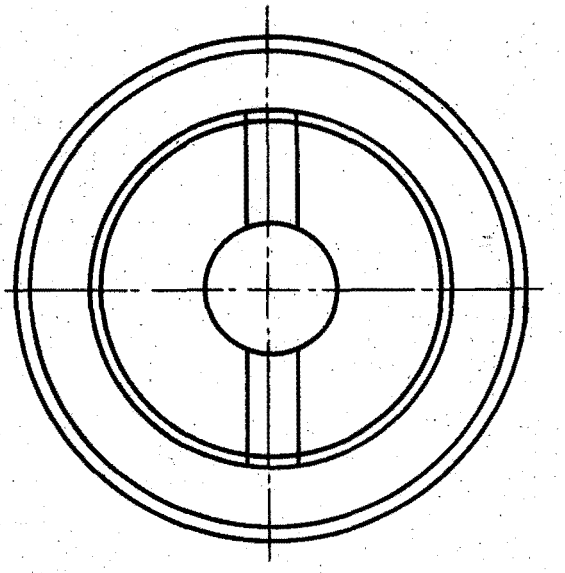
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81-3-37



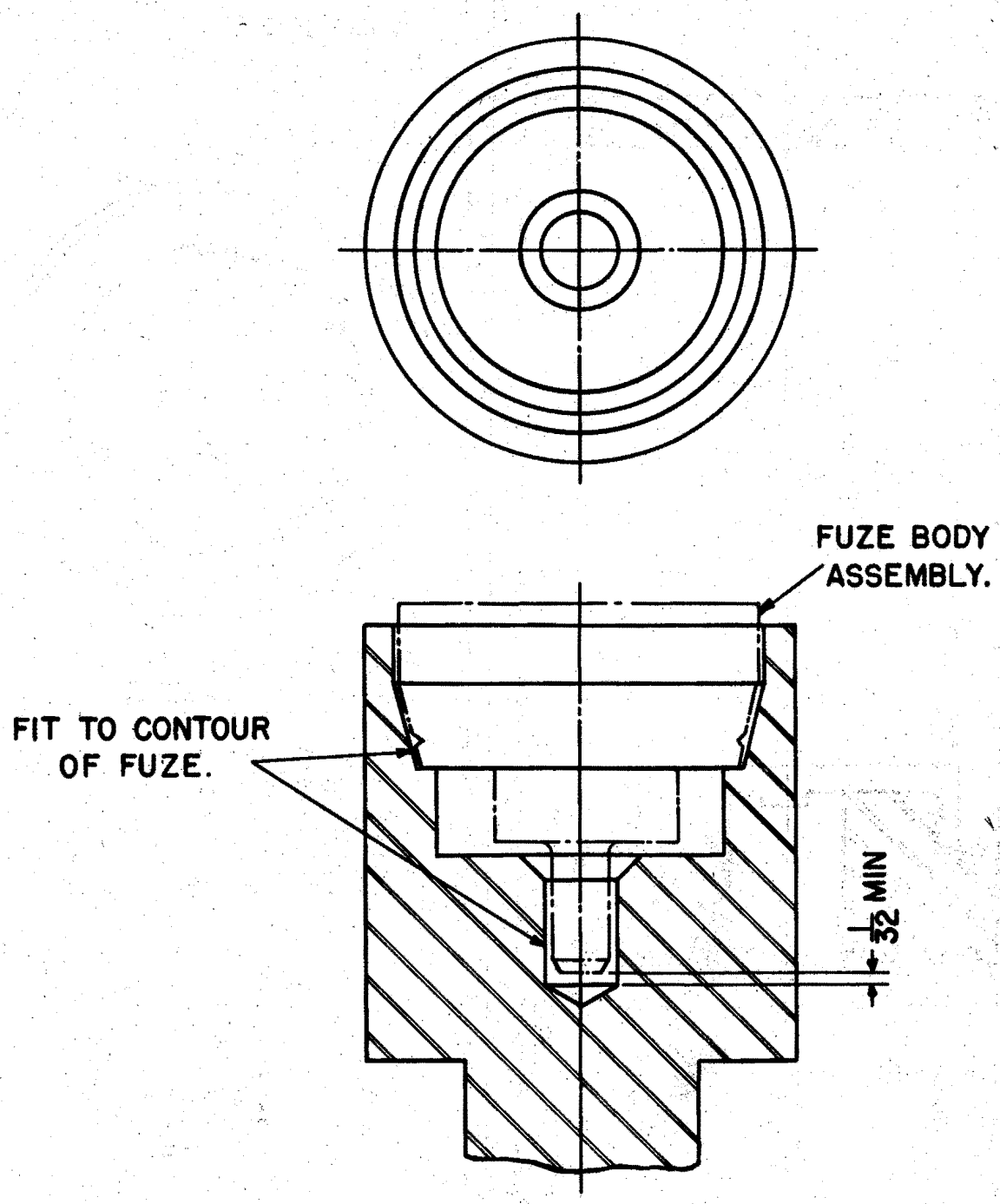
JUMBLE FIXTURE



ADAPTER
STEEL

FIGURE 2

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SPINDLE
TOOL STEEL, HARDEN

FIGURE 1

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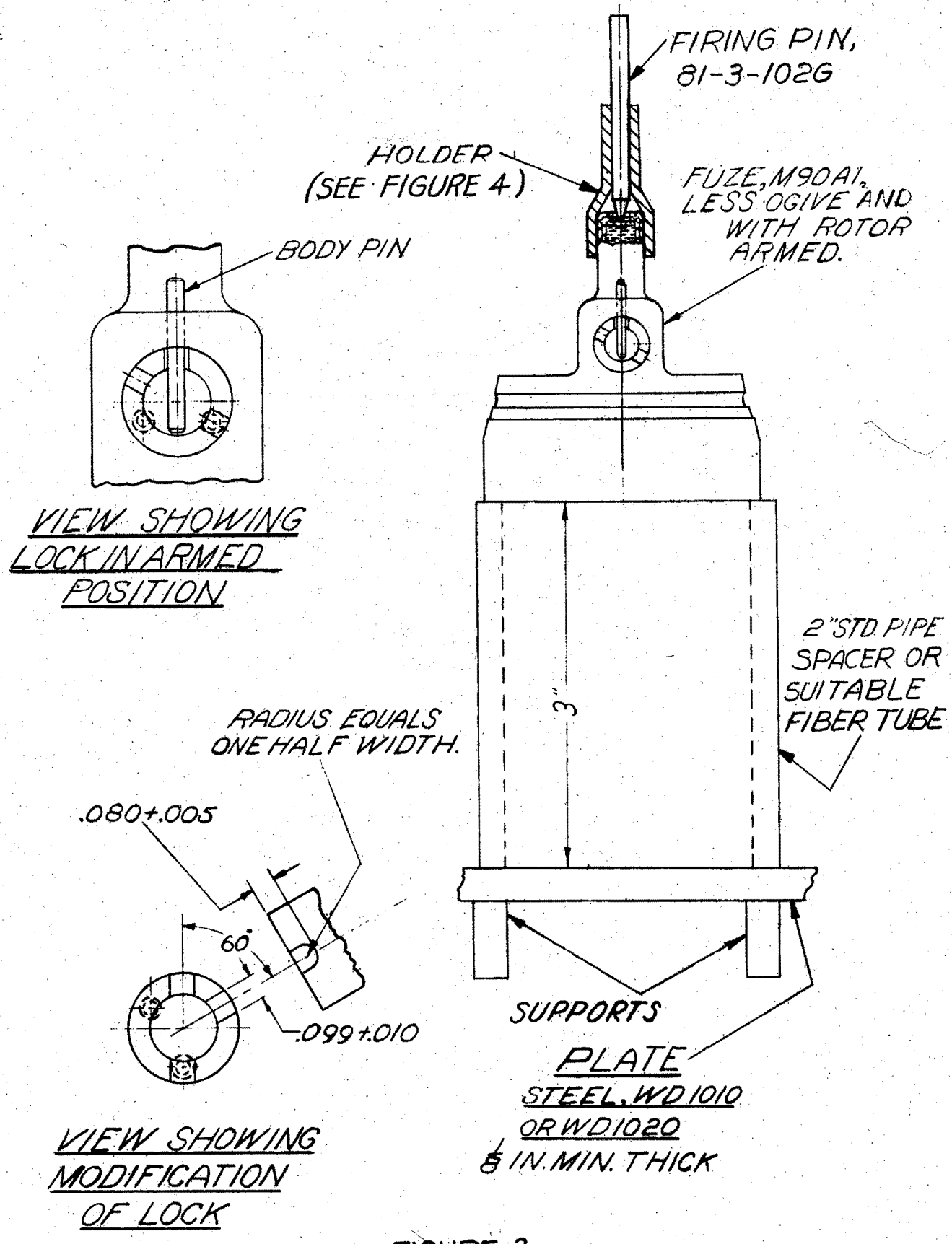
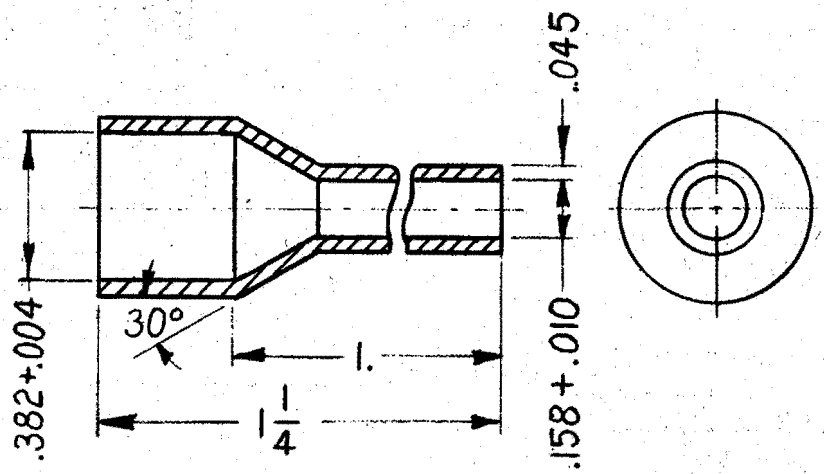


FIGURE 3

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HOLDER

PLASTIC, ALUMINUM OR
GILDING METAL
DESIGN NOT MANDATORY

FIGURE 4

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